Notes 5

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Linux Commands

1. ls

Definition: The `ls` command is used to show files and folders inside a directory.

Usage: `ls [options] [directory]`

Examples:

To show files in the current directory:

```
ls
```

To show hidden files:

```
ls -a
```

To show more details about files:

```
ls -l
```

2. pwd

Definition: The `pwd` command means print working directory. It shows where I am currently located in the terminal.

Usage: `pwd`

Examples:

To check my current directory:

```
pwd
```

To go to my home directory and check the location:

```
cd ~
pwd
```

3. cd

Definition: The `cd` command means change directory. It is used to move from one folder to another.

Usage: `cd [directory]`

Examples:

To go to the Documents folder:

```
cd Documents
```

To go back one directory:

```
cd ..
```

To go to my home directory:

```
cd ~
```

To go to the root directory:

```
cd /
```

Questions

1. What is a variable?

A variable is something that stores information in Linux. It can store a word, number, or other value.

Example:

```
name="John"
echo $name
```

2. How do I use a variable?

To use a variable, I give it a name and a value. Then I use `$` to get the value stored in it.

Example:

```
age=20
echo $age
```

This will show `20` in the terminal.

3. What is an environment variable?

An environment variable stores information about the system or shell. Programs can also use these variables.

Some examples are:

- `$HOME` - Shows my home directory.
- `$USER` - Shows my username.
- `$PATH` - Shows where Linux looks for commands.

Example:

```
echo $HOME
```

4. What is a user defined variable?

A user defined variable is a variable that I create myself to store information in the terminal.

Example:

```
color="blue"
echo $color
```

This will print `blue`.

5. What is the root directory?

The root directory is the main directory in Linux. It is represented by `/`. All other directories are located under the root directory.

Example:

```
cd /
ls
```

6. What does Parent Directory mean?

A parent directory is the folder that is one level above the current folder. In Linux, we use `..` to go to the parent directory.

Example:

```
cd ..
```

7. What does Current Working Directory mean?

The current working directory is the folder I am currently in when using the terminal.

I can use the `pwd` command to see my current location.

Example:

```
pwd
```

8. What is an absolute path? Include an example.

An absolute path is the full path to a file or folder. It always starts from the root directory `/`.

Example:

```
cd /home/student/Documents
```

This takes me directly to the Documents folder of the student user.

9. What is a relative path? Include an example.

A relative path is a path that starts from the directory I am currently in. It does not start with `/`.

Example:

```
cd Documents
```

If I am in my home directory, this command takes me to the Documents folder.

10. What is the difference between Your home directory and The home directory?

Your home directory is the personal folder for your user account. This is where your files and folders are stored.

For example:

```
/home/student
```

The `/home` directory is a folder that contains the home directories of regular users.

For example:

```
/home
```

The difference is that `/home` can contain many users' folders, but my home directory is only for my account.

I can go to my home directory by typing:

```
cd ~
```